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Endterm Progress Report: Literature Review and Conceptual Foundations

1. Linear Algebra & Geometric Transformations

In my review of multi-dimensional medical image computing, I focused on how spatial modifications are modeled mathematically as vectors and matrices.

Core Conceptual Takeaways

- Vector Spaces & Bases: I reviewed how a digital 3D volume is sampled on a discrete spatial grid, where any given voxel position is mapped as a position vector. Linear combinations of coordinate basis vectors allow us to navigate these multi-dimensional coordinate systems.
- Linear Transformations: I focused on formulating operations like scaling, rotation, and shearing as linear mappings, represented mathematically through matrix-vector multiplications (y = Ax).
- Determinants & Eigenspaces: I explored how the determinant evaluates the spatial scaling factor of a transformation. Tracking eigenvalues and eigenvectors allows us to isolate the invariant directions of structural mappings, which is a foundational requirement in tensor imaging modalities like Diffusion Tensor Imaging (DTI).

2. Probability Foundations & Bayesian Inference

Because clinical scans are inherently corrupted by statistical noise and artifacts, I studied probabilistic modeling to understand how image processing algorithms perform inference under uncertainty.

Core Conceptual Takeaways

- Probability Spaces & Conditional Probability: This framework allowed me to quantify the probability of a underlying clinical state given a set of observed pixel intensities, relying on joint distributions and sample space definitions.
- Probability vs. Likelihood: I clarified a major point of confusion regarding a fixed model with parameters theta and observed data x:
    - Probability evaluates p(x theta) to predict unknown outcomes x using a known, fixed parameter set theta.

    - Likelihood evaluates $L(\theta \mid x) = p(x \mid \theta)$, shifting the distribution parameters theta to determine which configuration most plausibly explains a fixed set of already observed data x.
- Bayesian Inference & MAP: I studied how Bayes' Theorem integrates prior clinical knowledge with newly acquired data:

$$p(\theta \mid x) = \frac{p(x \mid \theta)\, p(\theta)}{p(x)}$$

Maximum A Posteriori (MAP) estimation allows us to optimize this relationship, combining the data likelihood $p(x \mid \theta)$ with a prior probability distribution $p(\theta)$ to discover the most statistically probable parameter set.

3. Spatial Domain Convolution & Image Filtering

I investigated convolution as the primary spatial neighborhood mechanism used to denoise, smooth, and extract structural features from digital image grids.

Core Conceptual Takeaways

- Mathematical Convolution: I reviewed how a spatial bounding kernel K of size $M \times N$ slides over a 2D discrete image array I, computing a localized weighted sum:

$$(I * K)(x, y) = \sum_{m=-M/2}^{M/2} \sum_{n=-N/2}^{N/2} I(x - m, y - n)\, K(m, n)$$

- Spatial Neighborhood Averaging:
    - Mean/Box Filter: This uses uniform unweighted coefficients ($K_{m,n} = \frac{1}{M \cdot N}$). It acts as a low-pass filter that attenuates high-frequency noise but introduces heavy blurring artifacts across sharp structural edge boundaries.
    - Gaussian Filter: This assigns spatially decaying weights based on an isotropic Gaussian distribution governed by a standard deviation σ. This allows me to smooth out noise while preserving structural boundaries much more effectively than a standard box filter.

4. Frequency Domain Representation & Fourier Basics

I studied the Fourier Transform to understand how spatial images can be completely decomposed into an infinite sum of sinusoidal waves of varying frequencies, phases, and orientations.

Core Conceptual Takeaways

- The Fourier Transform: This mathematical tool maps a spatial image function $f(x, y)$ into its corresponding frequency spectrum representation $F(u, v)$:

$$F(u,v) = \iint block_{\infty}^{\infty} f(x,y)\, e^{-i2\pi(ux+vy)}\, dx\, dy$$

- Clinical Significance: This concept is deeply tied to Magnetic Resonance Imaging (MRI). MRI scanners do not sample spatial voxels directly; instead, the physical hardware measures signal phase and frequency over time, mapping directly to a raw frequency grid known as k-space. To reconstruct a viewable image, I learned that we must compute the Inverse Fast Fourier Transform (IFFT) of this k-space data.

5. Medical Image Input/Output (I/O) & Spatial Coordinates

I dedicated time to understanding how specialized clinical data files store multidimensional voxel matrices alongside vital geometric acquisition metadata.

Core Conceptual Takeaways

- File Formats: I noted that clinical environments avoid lossy compression formats (like JPEG or PNG) to protect data fidelity. Instead, they rely on specialized formats like NIfTI (.nii, .nii.gz) or DICOM.
- The Affine Matrix: I focused heavily on the affine transformation matrix stored inside image headers. This matrix maps discrete voxel array indices (i, j, k) directly to real-world physical anatomical coordinates (x, y, z) in millimeters:

$$\begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{32} & r_{32} & r_{33} & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} i \\ j \\ k \\ 1 \end{bmatrix}$$

This coordinate metadata is crucial because it accounts for the physical scaling, rotation, and translation transformations set by the scanner relative to the patient's orientation.

6. Regularization Norms & Sparsity Preview

Finally, I explored regularizers to see how inverse problems in medical imaging (such as reconstructing a full CT scan from highly under-sampled, low-radiation projection data) can be mathematically solved when they are ill-posed.

Core Conceptual Takeaways

- Objective Function Optimization: I analyzed how algorithms handle ill-posed systems by adding a regularization penalty parameter to a standard least-squares data loss term:

$$\min_{\theta} \left(\|X\theta - y\|_2^2 + \lambda \|\theta\|_p \right)$$

Norm Formulations (L2 vs. L1):

- Ridge Regularization (L2-norm, theta2^2 = sum theta_i^2): This penalizes large model coefficients uniformly, spreading weights smoothly across correlated variables to stabilize the solver.
- Lasso Regularization (L1-norm,theta_1 = sum theta_i): This forces uninformative coefficients to drop to exactly zero. This creates sparse models where only the most critical structural features are preserved.

7. Contextual Framework: Data Science Paradigms

Medical Image Computing (MIC) is situated within the broader discipline of Data Science, an interdisciplinary domain that unifies statistics, data analysis, machine learning, and optimization to understand and analyze real-world phenomena.

The Foundational Skillset

As a technical discipline, data science sits at the intersection of several fundamental pillars:

- Computer Science / Hacking Skills: Essential mechanics for acquiring, cleaning, manipulating, and scaling massive arrays of electronic records and big data.
- Math and Statistics Knowledge: Provides the theoretical rigor necessary to select appropriate tools, formulate hypotheses, and extract mathematically sound insights.
- Substantive Field Expertise: Critical domain-specific knowledge required to motivate clinical questions and accurately interpret algorithmic outcomes.

> The Danger Zone: Combining advanced programmatic "hacking" skills with field-specific domain knowledge—while bypassing a rigorous foundation in mathematical methods and statistics—creates a distinct analytical danger zone that frequently produces flawed or incorrect analyses.

The Evolution of Scientific Inquiry

According to Turing Award winner Jim Gray, data-intensive computing marks the arrival of a Fourth Paradigm in scientific discovery:

1. Experimental Science: Direct observation and empirical documentation of physical events.
2. Theoretical Science: Analytical formulations, physical laws, and mathematical equations.
3. Computational Science: Direct algorithmic simulation of complex physical models.
4. Data-Driven Discovery: Computer-assisted automation where data exploration and hypothesis generation themselves rely directly on processing massive data repositories.

Under this paradigm, computers parse vast, complex datasets to flag mathematically abnormal patterns, allowing human domain experts to isolate, model, and explain the underlying biological mechanisms.

8.Definitional Scope & Interdisciplinary Nature of MIC

As formalized in the CS-736 curriculum, Medical Image Computing is an interdisciplinary field dedicated to developing computational and mathematical methods to solve problems involving medical images, driving both biological research and clinical care.

The course operates at the intersection of three major areas:

- Computer Science & Engineering: Delivers the primary implementation mechanics, focusing on robust algorithm design, statistical inference, data structures, and computational scaling.
- Mathematics & Physics: Physics provides the analytical frameworks to model the underlying scanner hardware, wave interactions, and signal measurements. Mathematics provides the formalisms needed to structuralize the raw data and learn statistical relationships.
- Medicine: Establishes the real-world application targets, identifying vital diagnostic, therapeutic, and surgical problems encountered in the clinic.

Academic Requirements

The theoretical components of CS-736 assume a clear academic background:

- Fundamental Prerequisites: Linear algebra, basic optimization frameworks, graph theory, data structures, and undergraduate-level programming.
- Advanced Focus: A comprehensive command of probability and mathematical statistics.
- Image Basics: Familiarity with foundational image-processing operations, including spatial convolution, neighborhood filtering, and Fourier transforms.

9. Structural Properties of Digital Images vs. Diagnostic Signals

A primary distinction must be made between 1D clinical diagnostic signals, continuous analog images, and discrete multi-dimensional digital arrays.

Signal Modalities (Non-Image)

Clinical signals such as the Electrocardiogram (ECG) or Electroencephalogram (EEG) map variations across unstructured grids or continuous time domains. For instance:

- ECG: Models a 1 to 1 scalar function tracking cardiac muscle voltage fluctuations relative to time.
- EEG: Measures electrical brain activity across a series of unstructured sensor coordinates over time.

Analog vs. Digital Images

- Analog Images: Captured on continuous photographic film negatives (e.g., historical X-ray or optical microscopy film). They contain no native spatial discretization or intensity quantization, and their spatial resolution is bounded entirely by the underlying film grain size.
- Digital Images: The focus of this course, where spatial information is digitized directly by design during acquisition through the physical layout of the scanner sensor array.

Mathematical Representation of Digital Images

A digital image is mathematically formalized as a multi-dimensional Cartesian array of numbers mapping discrete coordinates to scalar or vector intensities:

- 2D Grayscale Array : Maps spatial coordinates (x,y) to a scalar intensity value representing localized tissue properties.
- Pixels vs. Voxels: Discrete elements on a 2D Cartesian spatial grid are referred to as pixels (*picture elements*). When scaled to a 3D structural volume, individual array coordinates are defined as voxels (*volume elements*).
- Dynamic Volumes: Formulated on a structured regular grid across both space and time
- Colormaps: Functions that translate numeric matrices into visual color representations for human display purposes. For example, scalar values inside a structural MRI matrix are mapped to gray tones, while metabolic values inside a PET scan are projected as colored heatmaps; the underlying raw matrix data remains entirely scalar.

10. Primary Mathematical Tasks in MIC

While this course treats scanner hardware as a mathematical "black box" without analyzing the internal instrumentation physics, it relies heavily on high-level specs that govern how underlying tissue configurations are converted into data arrays.

The primary core tasks within MIC focus on principled algorithm design, structuring image computing as an automated solver for explicit optimization objectives:

A. Image Reconstruction

- Objective: Synthesizing human-viewable, spatially coherent anatomical arrays from unstructured or raw frequency data collected along scanner detector arrays.

- Core Mechanisms: Leverages data acquisition specs via localized spatial convolutions, neighborhood filters, and Fourier transforms.

B. Image Denoising and Restoration

- Objective: Mitigating structural degradation, random noise, and scanner artifacts introduced during clinical acquisition.
- Core Mechanisms: Uses mathematical and statistical models to characterize system-specific corruption. Algorithms minimize clear objective functions to exploit these degradation models, preserving true underlying anatomy.

C. Image Segmentation

- Objective: Delineating structural boundaries and partitioning a discrete digital matrix into distinct, clinically meaningful anatomical components, organs, or pathologies.
- Core Mechanisms: Operates via three primary mathematical methodologies:
    1. *Machine Learning:* Driven by formal Bayesian modeling and variational inference derivations computed from scratch.
    2. *Graph Theory:* Formulates discrete optimization routines and path cuts evaluated directly from scratch.
    3. *Deformable Contours:* Relies on the physical principles of partial differential equations (PDEs) to iteratively evolve structural boundaries.

D. Image Registration

- Objective: Geometrically aligning and warping different multi-dimensional images into a unified coordinate space.
- Core Mechanisms: Requires mathematically defining statistical image similarity measures (e.g., geometry-driven or joint-intensity features), establishing the degrees of freedom for spatial warps, and employing robust optimization solvers.

E. Shape and Classification Analysis

- Shape Analysis: Statistically quantifies variations in human structures, defining normal vs. abnormal morphological configurations to aid in reconstructive surgeries or study disease progression.
- Image Classification: Employs predictive classifiers (e.g., Random Forests or Deep Learning networks) to parse localized spatial patterns—such as color, edge density, and texture—to automatically classify tumor lines, index cancer grades, or stage tissue spread.